

Emaan Heidari

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EDUCATION

University of Southern California - GPA: 3.97/4.00

Expected Dec 2027

B.S./M.S. in Computer Engineering and Computer Science (Integrated)

Los Angeles, CA

- Coursework: Parallel Computing, Algorithms, Data Structures, Machine Learning, Operating Systems, Distributed Systems, Computer Systems, Embedded Systems, Digital Design, Networking, Linear Algebra, Probability/Statistics

EXPERIENCE

Tesla

May 2026 – Present

Software Engineering Intern

Palo Alto, CA

- Building on-device systems software for Tesla Energy products using Rust and Go.

Liquid Propulsion Lab

Sep 2025 – Present

Software Engineer

Los Angeles, CA

- Architected a multithreaded real-time C++ data acquisition pipeline ingesting ~18k samples/sec across 3 LabJack devices and 38 sensors for a liquid bipropellant rocket engine test stand
- Implemented a low-latency UDP/TCP POSIX-socket pipeline over Ethernet in C++ with InfluxDB logging

SiFly

May 2025 – Sep 2025

Software Engineering Intern

Santa Clara, CA

- Implemented a 3D drone obstacle avoidance pipeline in Python on ROS2, integrating VFH+ local planning with PX4 offboard control via uXRCE-DDS at 16 Hz, achieving 97% collision-free navigation in Gazebo simulation
- Extended PX4's C++ flight stack with new failsafe modes for autonomous mission resumption after control loss
- Shipped a TypeScript/React monitoring dashboard for flight telemetry, mission status, and SQL-based log search

Tandem Launch

May 2024 – Aug 2024

Software Engineering Intern

Montreal, Canada

- Optimized CNN accelerator simulation throughput $2.2\times$ via NumPy vectorization, batching, and memory layout.
- Developed TensorFlow QAT layers to model 8-bit analog hardware constraints within 0.5% RMSE of the accelerator
- Cut hardware-in-the-loop firmware testing cycles 90% by designing a Docker-based CI infrastructure on Linux

PROJECTS

MurmurMatch.com | *Python, GCP, Firebase, PostgreSQL, FastAPI, Next.js/React, Stripe*

[\[Website\]](#)

- Built and scaled a college social platform to 56,000+ users used by 1/2 of Dartmouth, 1/3 of Brown, 1/3 of Stanford, and 1/4 of Princeton students, converting 500+ users to a \$7.99/week subscription
- Architected load-balanced GCP infrastructure with autoscaling Managed Instance Groups to support demand

Efficient Decode Attention | *C++, CUDA, CMake, Nsight Compute, Google Test + Benchmark*

[\[PDF\]](#) [\[Code\]](#)

- Implemented a flash-decode CUDA kernel eliminating the softmax bottleneck, achieving $28\times$ throughput over a standard attention kernel (477 Gi/s vs. 17 Gi/s) on an NVIDIA A40
- Designed a paged KV cache manager in C++ eliminating fragmentation via fixed-size page allocation, supporting $4-8\times$ more concurrent sequences at a constant $1.8\times$ per-token throughput cost

Representational Regularization for LLM Ensembles | *Python, PyTorch, HuggingFace*

[\[PDF\]](#) [\[Code\]](#)

- Fine-tuned a 30-model DeepSeek-7B ensemble with a CKA diversity penalty, reducing inter-model similarity by 19%
- Trained a residual MLP ensemble router, outperforming the strongest 7B baseline by an average of 4.2 percentage points across 5 benchmarks (MMLU, GSM8K, HumanEval, etc.) at negligible additional inference cost

FPGA Neural Network Inference Accelerator | *Verilog, Vivado, Python, PyTorch*

[\[PDF\]](#)

- Built a pipelined FPGA neural inference accelerator in Verilog with $8.95\mu\text{s}$ latency and $3.43\mu\text{J}$ energy per inference
- Implemented fixed-point deterministic inference with LUT activations and QAT, reaching 95.1% accuracy

SKILLS

Languages: C++ (17/20), C, Python, Rust, Go, Java, SQL, Bash, TypeScript, JavaScript, Verilog

Tech: CUDA, CMake, Linux, Git, perf, gdb, Valgrind, Clang/LLVM, GCC, OpenMP, MPI, Pthreads, PyTorch,

TensorFlow, Nsight Compute, Spark, NumPy, Docker, Grafana, AWS, GCP, FastAPI, Zephyr, ROS2, PX4, PostgreSQL